

TAEK LIM

M.S. Computer Science Student — UTRGV | Seeking Faculty Research Collaboration

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M.S. in Computer Science student with experience across infrastructure, backend, and frontend development, specializing in data processing. Interested in contributing to data-driven research and applying AI/ML techniques to analyze data and enhance existing research and systems.

EDUCATION

- M.S. Computer Science — The University of Texas Rio Grande Valley (Expected Dec 2027)
- M.S. Computer Science, Cybersecurity & Information Assurance — University of Central Missouri (May 2026)
- B.S. Mathematics-Economics — University of California, Los Angeles (Mar 2013)
- Graduate coursework: Advanced Operating Systems, Advanced Computer Networks, Advanced Algorithms, AI, Neural Networks and Deep Learning, Design of Cryptographic Algorithms and Protocols, Compiler Design, Cybersecurity
- Current coursework (Fall 2026): Foundations of Intelligent Security Systems, Bioinformatics, Introduction to Data Science

TECHNICAL EXPERTISE

- Languages: JavaScript/TypeScript, Java, Python, C/C++, SQL
- AI/ML: PyTorch, TensorFlow, LangChain/LangGraph, prompt engineering, applied LLM tooling (Claude, OpenAI, Gemini)
- Distributed Systems & Data: Kafka, TimescaleDB, gRPC/Protobuf, k3s, PostgreSQL/MySQL, Redis, Elasticsearch
- Cloud/DevOps: AWS (EC2/S3/RDS), Azure, Docker

RESEARCH-RELEVANT PROJECTS

- Weather Temperature Forecasting Platform (2026): Built a Kafka streaming pipeline with TimescaleDB storage; deployed a hybrid k3s cluster across cloud and local nodes; compared ML/DL models for forecasting accuracy.
- MapReduce Infrastructure Implementation (2024): Re-implemented a simplified distributed MapReduce system in C++/gRPC, based on the original Dean & Ghemawat paper.
- 5+ years of industry experience building large-scale, data-intensive systems from collecting to visualization, including a recommendation solution processing millions of daily user events and multi-channel marketing data analytics and visualization.

HOW I CAN CONTRIBUTE

- Systems and infrastructure work: building and maintaining data pipelines, distributed systems, and cloud deployments for research projects
- Software engineering support: designing APIs, backend services, and dashboards to support data collection and analysis
- Applied AI/ML implementation: prototyping and evaluating ML/DL models using PyTorch/TensorFlow
- Building research or lab websites, and creating data visualizations to communicate results
- Leveraging a math/econ, computer science, and practical software engineering background to participate in research, with strong self-directed learning ability

Interested faculty are welcome to reach out at taek.lim01@utrgv.edu to discuss potential collaboration.